Finite-Time Centroid Tracking and Exponential Formation Stabilization in Arbitrary Swarm Configurations

Yash Purswani*, Shivansh Agarwal* and Anuj Tiwari

***Abstract*— Certifying finite arrival times and convergence rates is essential for coordinating swarms in time-critical missions. However, these convergence properties generally depend on the interaction graph and feedback gains in existing methods, leaving exact acceptance deadlines uncertified for arbitrary time-varying formations. To address this limitation, this letter proposes a kinematic swarm model in which centered, time-varying stations are prescribed through individually assigned offsets with feedforward, driven by a shared finite-time centroid tracking feedback. It is shown that this shared feedback is dissipative for the relative formation error, which decouples the mission-level and agent-level channels in closed form. As a result, the swarm centroid is certified to track the reference in finite time, and individual station errors decay globally and exponentially from the initial condition. This decoupling also provides an explicit deadline certifying when every agent enters an arbitrary tolerance ball. Furthermore, the formation guarantee extends to limited sensing across switching proximity components. Numerical simulations confirm both certificates across rotating and reconfiguring formations.**

***Index Terms*— Control of networks, cooperative control, stability of nonlinear systems.**

I. Introduction

Multi-robot teams routinely perform tasks difficult for individual platforms: aerial swarms map expansive terrain [1], warehouse fleets sort inventory in parallel [2], and aerial manipulators jointly transport heavy loads [3]. Such a deployment is only useful once the team behaves as one, following a common mission trajectory while jointly holding a prescribed geometric arrangement – the baseline of a distributed sensor [4], the pose of a jointly carried load, or the corridor a team occupies through a confined space. Such arrangements are frequently irregular rather than symmetric, so a swarm model restricted to symmetric or graph-emergent shapes cannot represent them.

In a growing share of these missions, the arrival is time-critical: a docking maneuver has a window, a handoff or a grasp is safe only once every agent is simultaneously within tolerance, and an intercept fails outright if late. Such a guarantee must be computable before deployment, from the initial condition alone, and must hold for each agent individually in addition to the group average. Finite-time stability [5], [6] certifies exact arrival with a time bounded by the initial state, and has been incorporated into consensus and formation designs [7], [8].

Formation control and trajectory tracking are each mature (Section II), but most schemes that combine them merge the reference and the formation shape into a single consensus-type error. Feasibility, convergence rate, and arrival time then all depend on the interaction graph, agent count, and gains; tracking and per-agent convergence are not certified independently, and existing finite-time results inherit the same dependence.

Fedele and D'Alfonso [9] proposed a kinematic swarm model in which a single fractional-power correction on the group centroid renders its tracking finite-time, independently of the interaction. The model has two limitations that matter here, i) the desired formation is not independently prescribed, but emerges from the pairwise interaction, so shapes that are not consistent equilibria of the update law in [9] cannot be realized; and, ii) only a swarm level convergence guarantee of the centroid is provided without individual-agent settling guarantee, since the agent formation error w.r.t. its desired position in the swarm is bounded only for $t \geq \tau_c$, after the centroid has settled, leaving the preceding transient uncertified.

This letter develops a kinematic swarm model that closes both gaps. Each agent is assigned an independent, possibly time-varying offset with feedforward about the moving centroid, subject only to the centering condition that the offsets sum to zero, so any centered formation – symmetric or not – is realizable and reconfigurable at will. The corrector remains common to the swarm and acts only on the centroid channel; it is shown to be dissipative for the formation error as well, which decouples the group- and agent-level channels in closed form and yields an individual-agent convergence certificate holding from the initial instant. The contributions are:

1) a kinematic model with individually assigned, centered C^1 offsets, so an arbitrary, asymmetric and time-varying formation is prescribed rather than emergent (Section III-A);
2) finite-time centroid tracking with τ_c set by $\sigma(t_0)$ and β alone, and not by M, ν, ε, n or $p_i(\cdot)$ (Theorem 1);
3) global exponential convergence to the stations from t_0, at a rate independent of M (subject only to $M+M^\top \succeq 0$), ν, ε, n and the formation geometry (Theorem 2);
4) a closed-form acceptance deadline T_{dock} for any tolerance, computable at t_0 from $\|\delta(t_0)\|$ (Corollary 1);
5) an extension to limited sensing, in which both formation results hold across any split of the proximity graph, which becomes complete after a computable time once the sensing radius exceeds the formation diameter (Section III-D).

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II. Background and Problem Statement

This section provides the relevant background on current state of formation control, finite time convergence literature, kinematic swarm models in the literature, which motivates the problem statement.

Formation control. Classical formation strategies rely on position-, displacement-, distance-, or bearing-based feedback [10], [11], virtual structures [12], rigid frameworks [13], or Laplacian maneuvers [14]. In each case, geometric shape and reference tracking are fundamentally coupled, leaving both the admissible formations and their convergence rates subordinate to the underlying interaction network.

Finite-, fixed-, and prescribed-time coordination. Translating convergence rates into operational deadlines has driven substantial research. Finite-time consensus bounds depend directly on initial disagreement and algebraic connectivity [7], while fixed-time methods [15]–[17] decouple the bound from initial states but retain scaling dependencies on network size, connectivity, and feedback gains. Prescribed-time controllers [18], [19] assign exact settling times via time-varying gains, at the expense of terminal singularities on $[t_0, T)$ and elevated noise sensitivity. An explicit arrival certificate derived strictly from initial conditions via time-invariant feedback remains absent.

Swarm kinematic models. One quantity does escape the interaction. Building on the aggregation analysis of Gazi and Passino [20], Fedele and D'Alfonso [9] drive the centroid with a common finite-time corrector, alongside a distance-weighted interaction ruled by a coupling matrix M, making the centroid's arrival interaction-independent; extensions generalize M to rotating, mixed, discrete- and fixed-time variants [21], [22]. The formation itself, however, is still whatever the interaction produces, and the agent-level guarantee is a hyper-ball of radius set by M, ν, ε and n, entered only once the centroid has settled.

The existing approaches do not provide independent certificates for swarm-level tracking and agent-level formation keeping objectives simultaneously, which leads to the main research objectives for the current work as follows,

1) *Arbitrary formation prescription.* Specify an arbitrary, time-varying and asymmetric formation independently of the interaction graph and coupling matrix, rather than letting it emerge from the interaction;
2) *Agent-level finite-time certification.* Certify, for every agent individually from t_0, an explicit deadline for entering an arbitrary tolerance of its station, independently of the interaction graph, the coupling gains and the swarm size.

III. Proposed Approach

This section develops the kinematic model that addresses both problems above: Section III-A defines the model together with the assigned, centered stations; Section III-B recovers the finite-time centroid guarantee under the new model; and Section III-C establishes the corrector's dissipativity for the formation error, from which the agent-level certificates follow; Section III-D relaxes the all-to-all sensing assumption.

A. Kinematic Model

Consider $n \geq 2$ agents in $\mathbb{R}^d$ with positions $x_i(t)$, $t \geq t_0$, and centroid $c(t) := \frac{1}{n}\sum_{i=1}^{n} x_i(t)$; every agent knows the position of every other, an assumption relaxed in Section III-D. The kinematic model of [9] is extended by assigning each agent a time-varying offset $p_i(t)$ together with its feedforward $\dot{p}_i(t)$:

$$\dot{x}_i = -(x_i - p_i - r) + \dot{r} + \dot{p}_i - \zeta - \frac{M}{n}\sum_{j=1}^{n} \underbrace{\frac{(x_i - p_i) - (x_j - p_j)}{\|x_i - x_j\|^{\nu} + \varepsilon}}_{=:\,\Gamma_{ij}}, \quad (1)$$

where

- $r \in C^1$ is the reference for the centroid, and $\sigma := c - r$ is the centroid error;
- $p_i \in C^1$ places agent i's *assigned station* at $p_i(t) + r(t)$;
- $\zeta_s = \operatorname{sign}(\sigma_s)\,|\sigma_s|^{\beta}$, $s = 1, \ldots, d$, $\beta \in (0, 1)$, is the finite-time corrector, common to all agents;
- $M \in \mathbb{R}^{d\times d}$ with $M + M^{\top} \succeq 0$ (i.e. $v^{\top} M v \geq 0$ for all $v \in \mathbb{R}^d$; M need not be symmetric), $\nu \in \mathbb{N}^{+}$ and $\varepsilon > 0$ set the coupling gain, its decay with distance, and its regularization;
- the coupling term Γ_{ij} is skew-symmetric: its denominator $D_{ij} := \|x_i - x_j\|^{\nu} + \varepsilon$ is symmetric in (i, j) and its numerator is skew-symmetric, so

$$\Gamma_{ij} = -\Gamma_{ji}. \quad (2)$$

For $p_i \equiv 0$, Eq. (1) is exactly the model of [9, eq. (1)]. The offsets are constrained only by the following assumption.

Assumption 1 (Centering)**.** *The offsets satisfy*

$$\sum_{i=1}^{n} p_i(t) = 0_d, \qquad \forall t \geq t_0. \quad (3)$$

Since Eq. (3) holds identically in t, it implies $\sum_{i=1}^{n} \dot{p}_i(t) = 0_d$ without further assumption, and it places the mean station $\frac{1}{n}\sum_i (p_i + r)$ exactly on r.

B. Trajectory Tracking

Theorem 1 is based on [9, Th. 1] with minor adaptations, Assumption 1 removing the assigned offsets from the centroid dynamics.

Theorem 1 (Finite-Time Centroid Convergence [9])**.** *Under Assumption 1, the centroid error of Eq.* (1) *satisfies* $\sigma(t) \equiv 0_d$ *for all* $t \geq \tau_c$*, where*

$$\tau_c = t_0 + \max_{s=1,\ldots,d} \frac{|\sigma_s(t_0)|^{1-\beta}}{1-\beta}, \quad (4)$$

independent of M, ν, ε *and of* $p_i(\cdot)$.

Proof. Averaging Eq. (1) over $i = 1, \ldots, n$,

$$\dot{c}(t) = -\frac{1}{n}\sum_i (x_i - p_i) + r + \dot{r} + \frac{1}{n}\sum_i \dot{p}_i - \zeta - \frac{M}{n^2}\sum_i\sum_j \Gamma_{ij}.$$

By the centering condition Eq. (3), $\frac{1}{n}\sum_i (x_i - p_i) = c(t)$ and $\frac{1}{n}\sum_i \dot{p}_i = 0_d$; and by the skew-symmetry Eq. (2), $\sum_i \sum_j \Gamma_{ij}(t) = 0_d$. Hence $\dot{c} = -c + r + \dot{r} - \zeta$, and since $\sigma := c - r$,

$$\dot{\sigma}(t) = -\sigma(t) - \zeta(t). \tag{5}$$

Since ζ_s depends only on σ_s, Eq. (5) represents d independent scalar equations, one per coordinate s,

$$\dot{\sigma}_s(t) = -\sigma_s(t) - \operatorname{sign}(\sigma_s(t))|\sigma_s(t)|^{\beta}, \quad s = 1, \dots, d. \tag{6}$$

For each s, consider the Lyapunov function $V_s(t) := \frac{1}{2}\sigma_s(t)^2$, the derivative of which along Eq. (6) is exactly

$$\begin{aligned}\dot{V}_s(t) &= \sigma_s \dot{\sigma}_s = -\sigma_s(t)^2 - \sigma_s(t)\operatorname{sign}(\sigma_s(t))|\sigma_s(t)|^{\beta} \\ &= -\sigma_s(t)^2 - |\sigma_s(t)|^{1+\beta}.\end{aligned} \tag{7}$$

Dropping the non-positive term $-\sigma_s(t)^2$ gives the bound

$$\dot{V}_s(t) \le -|\sigma_s(t)|^{1+\beta} = -2^{\eta} V_s(t)^{\eta}, \tag{8}$$

where the last equality rewrites $|\sigma_s|^{1+\beta}$ in terms of V_s, using $|\sigma_s| = \sqrt{2V_s}$, and $\eta := \frac{1+\beta}{2} \in (\frac{1}{2}, 1)$. Finally, by the standard finite-time comparison argument [6], separating variables and integrating (8) from t_0 to t gives

$$\begin{cases} V_s^{1-\eta}(t) \le V_s^{1-\eta}(t_0) - 2^{\eta}(1-\eta)(t - t_0), & t \in [t_0, \tau_s] \\ V_s(t) = 0, & \forall t \ge \tau_s, \end{cases} \tag{9}$$

where t_0 is the initial time and τ_s — the instant at which the bound in Eq. (9) reaches zero — is

$$\tau_s = t_0 + \frac{V_s^{1-\eta}(t_0)}{2^{\eta}(1-\eta)} = t_0 + \frac{|\sigma_s(t_0)|^{1-\beta}}{1-\beta}. \tag{10}$$

Therefore, each component $\sigma_s(t)$ reaches zero in finite time, no later than τ_s. Setting $\tau_c := \max_{s=1,\dots,d} \tau_s$ then gives $\sigma(t) \equiv 0_d$ for all $t \ge \tau_c$, which is Eq. (4), independent of $M, \nu, \varepsilon, p_i(\cdot)$ since these were eliminated already at Eq. (5). □

C. Formation Tracking

Define the formation error of agent i, $\delta_i := x_i - p_i - r$, its offset from its assigned station, stacked as $\delta := [\delta_1^{\top}, \dots, \delta_n^{\top}]^{\top} \in \mathbb{R}^{nd}$. Since $(x_i - p_i) - (x_j - p_j) = \delta_i - \delta_j$, Eq. (1) gives

$$\dot{\delta}_i = -\delta_i - \zeta - \frac{M}{n}\sum_{j=1}^{n} \Gamma_{ij}, \qquad \Gamma_{ij} = \frac{\delta_i - \delta_j}{D_{ij}}. \tag{11}$$

Lemma 1 (Centroid–Formation Identity)**.** *Under Assumption 1, $\sum_{i=1}^{n} \delta_i(t) = n\,\sigma(t)$ for all $t \ge t_0$, and consequently*

$$-\sum_{i=1}^{n} \delta_i^{\top}(t)\zeta(t) = -n\sum_{s=1}^{d} |\sigma_s(t)|^{1+\beta} \le 0. \tag{12}$$

Proof. Summing $\delta_i = x_i - p_i - r$ over i and using the centering condition Eq. (3),

$$\sum_{i=1}^{n} \delta_i(t) = nc(t) - 0_d - nr(t) = n\sigma(t).$$

Since $\zeta(t)$ carries no agent index, it factors out of the sum,

$$\begin{aligned}\sum_{i=1}^{n} \delta_i^{\top}(t)\zeta(t) &= n\sigma^{\top}(t)\zeta(t) \\ &= n\sum_{s=1}^{d} \sigma_s(t)\operatorname{sign}(\sigma_s(t))|\sigma_s(t)|^{\beta} \\ &= n\sum_{s=1}^{d} |\sigma_s(t)|^{1+\beta},\end{aligned}$$

which gives Eq. (12). □

Theorem 2 (Global Exponential Formation Convergence)**.** *Under Assumption 1, the solutions of Eq.* (1) *satisfy $\|\delta(t)\| \le \|\delta(t_0)\|\, e^{-(t-t_0)}$ for all $t \ge t_0$, independent of M (subject only to $M + M^{\top} \succeq 0$), ν, ε, n and the formation geometry; consequently $\|\delta_i(t)\| \le \|\delta(t)\|$ for every i, and $\delta_i(t) \to 0_d$.*

Proof. Consider the Lyapunov function $V_\delta(t) := \frac{1}{2}\|\delta(t)\|^2 = \frac{1}{2}\sum_i \|\delta_i(t)\|^2$. Differentiating,

$$\dot{V}_\delta(t) = \sum_i \delta_i^{\top}(t)\dot{\delta}_i(t).$$

Substituting Eq. (11) and distributing $\delta_i^{\top}$,

$$\begin{aligned}\dot{V}_\delta(t) = &-\sum_i \|\delta_i(t)\|^2 - \sum_i \delta_i^{\top}(t)\zeta(t) \\ &- \frac{1}{n}\sum_i \sum_j \delta_i^{\top}(t) M \Gamma_{ij}(t).\end{aligned} \tag{13}$$

By Lemma 1, the second term equals $-n\sum_s |\sigma_s(t)|^{1+\beta} \le 0$. For the third, let $\Omega(t) := \sum_i \sum_j \delta_i^{\top}(t) M \Gamma_{ij}(t)$; relabeling $i \leftrightarrow j$ and using $\Gamma_{ji} = -\Gamma_{ij}$ gives $\Omega(t) = -\sum_i \sum_j \delta_j^{\top}(t) M \Gamma_{ij}(t)$, and averaging the two expressions,

$$\begin{aligned}\Omega(t) &= \frac{1}{2}\sum_i \sum_j \big(\delta_i(t) - \delta_j(t)\big)^{\top} M \Gamma_{ij}(t) \\ &= \frac{1}{2}\sum_i \sum_j \frac{\big(\delta_i(t) - \delta_j(t)\big)^{\top} M \big(\delta_i(t) - \delta_j(t)\big)}{D_{ij}(t)}.\end{aligned} \tag{14}$$

Since $M + M^{\top} \succeq 0$, $(\delta_i - \delta_j)^{\top} M (\delta_i - \delta_j) \ge 0$ (only the symmetric part of M enters this quadratic form), and $D_{ij} \ge \varepsilon > 0$, so $\Omega(t) \ge 0$ and the third term of Eq. (13) is non-positive. Discarding both non-positive terms,

$$\dot{V}_\delta(t) \le -\sum_i \|\delta_i(t)\|^2 = -2V_\delta(t). \tag{15}$$

Multiplying Eq. (15) by e^{2t} and using $\frac{d}{dt}[e^{2t}V_\delta(t)] = e^{2t}\big(\dot{V}_\delta(t) + 2V_\delta(t)\big)$,

$$\frac{d}{dt}\big[e^{2t}V_\delta(t)\big] \le 0,$$

so $e^{2t}V_\delta(t)$ is non-increasing; integrating from t_0 to t gives $V_\delta(t) \le V_\delta(t_0)e^{-2(t-t_0)}$, and since $V_\delta = \frac{1}{2}\|\delta\|^2$,

$$\|\delta(t)\| \le \|\delta(t_0)\|\, e^{-(t-t_0)}, \qquad \forall t \ge t_0. \tag{16}$$

Since $\|\delta_i(t)\|^2 \le \sum_k \|\delta_k(t)\|^2 = \|\delta(t)\|^2$ for every i, Eq. (16) gives $\|\delta_i(t)\| \le \|\delta(t)\| \le \|\delta(t_0)\|\, e^{-(t-t_0)}$, and letting $t \to \infty$ gives $\delta_i(t) \to 0_d$. □

Remark 1. *Compared to Theorem 1, note the difference in convergence type: σ reaches exactly zero at the finite time τ_c, since $\eta < 1$ in Eq.* (8) *keeps the correction strong all the way to the origin; here, once ζ's contribution is absorbed by Lemma 1, $\dot{V}_\delta \le -2V_\delta$ is a linear law, so $\delta(t)$ only approaches zero asymptotically.*

The bound holds from t_0: by Lemma 1 the corrector is dissipative for δ before the centroid settles, so neither Eq. (16) nor the deadline below waits for τ_c.

Corollary 1 (Practical Finite-Time Convergence)**.** *For any tolerance $\varepsilon_{\mathrm{tol}} > 0$ (e.g. a physical docking or gripper clearance margin), $\|\delta_i(t)\| \le \varepsilon_{\mathrm{tol}}$ for all i and all $t \ge T_{\mathrm{dock}}$, where*

$$T_{\mathrm{dock}} = t_0 + \max\Big(0,\ \ln \frac{\|\delta(t_0)\|}{\varepsilon_{\mathrm{tol}}}\Big). \tag{17}$$

Proof. By Theorem 2,

$$\|\delta_i(t)\| \le \|\delta(t)\| \le \|\delta(t_0)\|\, e^{-(t-t_0)}$$

for every agent i. Forcing the right-hand side below $\varepsilon_{\mathrm{tol}}$ requires

$$\|\delta(t_0)\|\, e^{-(t-t_0)} \le \varepsilon_{\mathrm{tol}} \quad \Longleftrightarrow \quad t - t_0 \ge \ln \frac{\|\delta(t_0)\|}{\varepsilon_{\mathrm{tol}}}$$

whenever $\|\delta(t_0)\| > \varepsilon_{\mathrm{tol}}$, and no wait is needed if $\|\delta(t_0)\| \le \varepsilon_{\mathrm{tol}}$; both cases together give Eq. (17). □

D. Extension to Limited Connectivity

Following [9, Sec. V-A], suppose each agent senses only within radius $R_s > 0$ and shares data within its connected component $\mathcal{G}_{q(i)}$ of size $n_q = |\mathcal{G}_q|$. With component centroid $c_q = \frac{1}{n_q}\sum_{i\in\mathcal{G}_q} x_i$ and mean station offset $\bar{p}_q = \frac{1}{n_q}\sum_{i\in\mathcal{G}_q} p_i$, the component tracking error is $\tilde{\sigma}_q = c_q - \bar{p}_q - r$. Each agent restricts its corrector and coupling to its component:

$$\dot{x}_i = -(x_i - p_i - r) + \dot{r} + \dot{p}_i - \zeta(\tilde{\sigma}_{q(i)}) - \frac{M}{n_{q(i)}} \sum_{j\in\mathcal{G}_{q(i)}} \Gamma_{ij}. \tag{18}$$

Both certificates survive this restriction because Lemma 1 applies component-wise: summing δ_i over $\mathcal{G}_q$ yields $\sum_{i\in\mathcal{G}_q}\delta_i = n_q\tilde{\sigma}_q$ for any partition without requiring Assumption 1. Provided the topology switches finitely often on bounded intervals, the following guarantees hold:

Formation invariance: Within each component, the corrector remains dissipative. Because $\delta(t)$ is continuous across component mergers and splits, Eq. (16) and deadline T_{dock} (17) hold unconditionally for all $t \ge t_0$. In the decoupled limit $\mathcal{G}_q = \{i\}$, (18) simply reduces to n independent finite-time station trackers.

Centroid tracking: On any dwell interval $[t_1, t_2)$ where components remain static, $\dot{\tilde{\sigma}}_q = -\tilde{\sigma}_q - \zeta(\tilde{\sigma}_q)$, certifying that each component error $\tilde{\sigma}_q$ vanishes by $t_1 + \max_s |\tilde{\sigma}_{q,s}(t_1)|^{1-\beta}/(1-\beta)$. Under Assumption 1, $\sigma = \frac{1}{n}\sum_q n_q\tilde{\sigma}_q$, ensuring the global centroid reaches the reference once all components settle.

Connectivity recovery: Let $D_p := \sup_{t\ge t_0}\max_{i,j}\|p_i(t) - p_j(t)\|$. Since $\|x_i - x_j\| \le \sqrt{2}\,\|\delta\| + D_p$, choosing $R_s > D_p$ guarantees the proximity graph becomes complete and remains connected for all $t \ge T_{\mathrm{conn}}$, where

$$T_{\mathrm{conn}} = t_0 + \max\Big(0,\ \ln \frac{\sqrt{2}\,\|\delta(t_0)\|}{R_s - D_p}\Big). \tag{19}$$

Beyond T_{conn}, Theorem 1 applies globally. Network connectivity therefore accelerates transient coordination through coupling and ensures a unified centroid, but is not required to certify agent-level station acceptance.

TABLE I

SIMULATION SETUP.

Parameter	Value
Agents, dimension (n, d)	10, 2
Formation template	ring, radius 5
Rotation rate ω (time-varying run)	0.5 rad/s
Reconfiguration (Fig. 2)	ring, wedge, star, arrow
Blend duration; start times	4 s; $t = 14, 26, 38$ s
Reference $r(t)$	$[t,\ 0]^\top$
Interaction matrix M	$5.0\, I_2$
Decay exponent ν	2
Regularizer ε	1
Sliding exponent β	0.5 (swept in Fig. 1)
Initial positions	$\mathcal{N}(0, 25I_2)$, seed 42
Integrator	LSODA, rtol 10^{-10}, atol 10^{-12}
Arrival event	$\lvert\sigma_s\rvert = 10^{-8}$, then latched
$\|\sigma(t_0)\|$	1.5762
$\|\delta(t_0)\|$	25.0657

TABLE II

PREDICTED SETTLING TIME EQ. (4) AND AVERAGED CONTROL EFFORT FOR EACH EXPONENT.

β	τ_c (s)	peak $\|u_{\mathrm{avg}}\|$	$\int \|u_{\mathrm{avg}} - \dot{r}\|\, dt$
0.2	1.51	2.49	1.594
0.5	2.25	2.56	1.580
0.8	5.24	2.63	1.576
1	∞	2.68	1.576

IV. NUMERICAL VALIDATION

To validate Theorems 1 and 2 and Corollary 1, the kinematic model Eq. (1) is integrated numerically and the resulting trajectory is compared against the analytical bounds Eq. (4), Eq. (16), and Eq. (17).

All experiments use the same swarm of $n = 10$ agents in $d = 2$ dimensions, started from the same initial positions and tracking $r(t) = [t,\ 0]^\top$ with the parameters of Table I. Three experiments are reported: the centroid channel for several exponents β (Section IV-A), the formation channel on a ring template under several couplings (Section IV-B), and a reconfiguration through four templates (Section IV-C).

A. Centroid Convergence and Its Control Cost

Fig. 1(a) shows $\|\sigma(t)\|$ for $\beta \in \{0.2, 0.5, 0.8\}$ and the linear baseline $\beta = 1$. Each fractional exponent reaches zero before its predicted τ_c (Table II), larger β arriving later, while the linear law only decays exponentially.

Fig. 1(b) shows the averaged command $\|u_{\mathrm{avg}}\| = \|\dot{c}\|$. Averaging Eq. (1) gives $\dot{c} = -\sigma + \dot{r} - \zeta(\sigma)$, since the

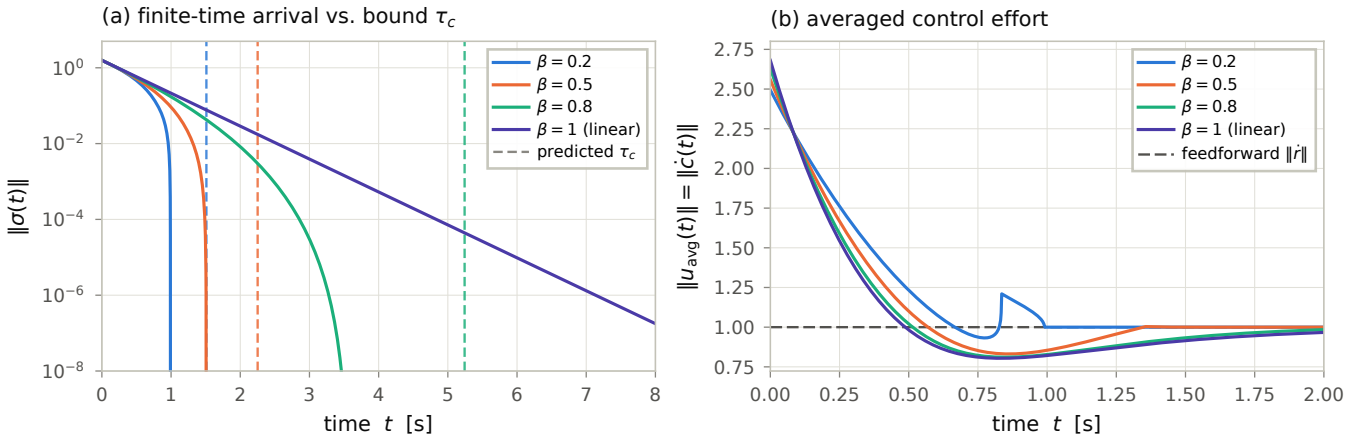


Fig. 1. Performance of the centroid channel across sliding exponents $\beta \in \{0.2, 0.5, 0.8\}$ alongside the linear baseline $\beta = 1$. (a) Convergence of the centroid tracking error $\|\sigma(t)\|$ plotted against the analytical settling-time bounds τ_c Eq. (4) (dashed lines). (b) Averaged control effort $\|u_{\text{avg}}(t)\| = \|\dot{c}(t)\|$ decaying to the feedforward baseline $\|\dot{r}\|$.

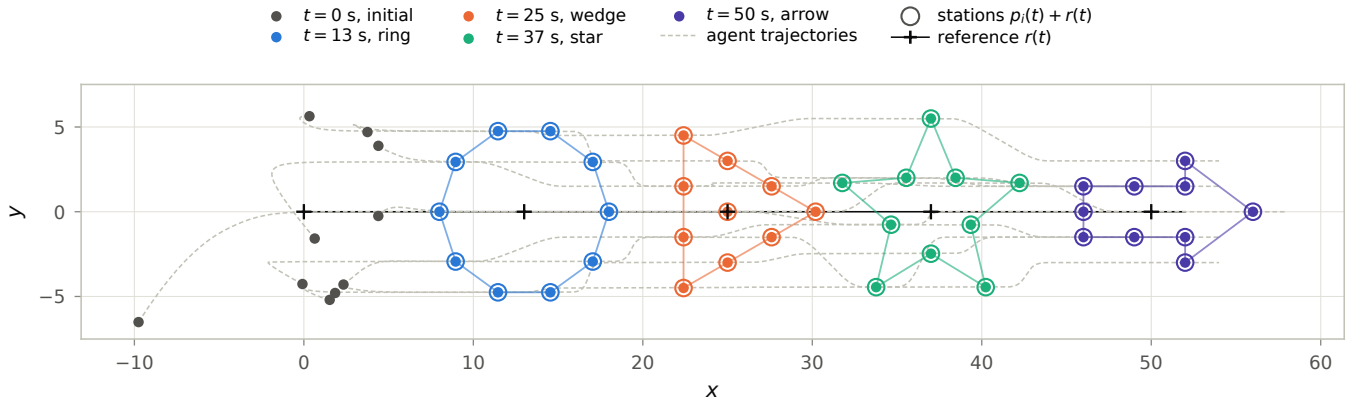


Fig. 2. Trajectory profiles for a swarm of $n = 10$ agents tracking a linear reference $r(t) = [t, 0]^\top$ while reconfiguring across four distinct centered templates: ring, wedge, star, and arrow. Dashed curves represent individual agent paths; hollow circles mark the time-varying target stations $p_i(t) + r(t)$ over template outlines, and filled dots indicate instant agent positions. Minimum-jerk blends over 4 s intervals preserve the centering condition $\sum p_i = 0$ throughout transitions, with agent-to-station slot assignment optimized via the Hungarian algorithm. A video of this simulation is available as drive link.

coupling cancels by Eq. (2) and the $\dot{p}_i$ by Eq. (3), so the effort depends on the centroid error alone: it is largest at t_0, decays as σ closes, and settles on the feedforward level $\|\dot{r}\|$ once the centroid arrives. It dips below that level while the corrector opposes the reference velocity, here because the centroid starts ahead of r, and the jumps mark the arrival of individual components σ_s. Since $|\sigma_s|^\beta < |\sigma_s|$ for $|\sigma_s| > 1$ and the reverse for $|\sigma_s| < 1$, a smaller β commands less while an error component exceeds one and more once it falls below, which is why the fractional laws start lower yet arrive first.. The corrective effort $\int \|u_{\text{avg}} - \dot{r}\|\, dt$ is nearly the same for every β (Table II): it is the length of the path of σ to the origin, essentially the distance $\|\sigma(t_0)\|$ whichever law moves it.

B. Formation Convergence and Acceptance Time

Table III shows the results on a ring formation template using Table I config under several couplings. They include a non-symmetric $M = 5I_2 + 5J$, with J the rotation by 90°, and the ring rotating rigidly at $\omega = 0.5$ rad/s, which exercises the $\dot{p}_i$ feedforward and preserves Assumption 1 since, with $R(t)$ the rotation by ωt, $\sum_i R(t)p_i = R(t)\sum_i p_i = 0_d$. For each run, $\hat{\lambda}$ is the largest λ with $\|\delta(t)\| \le \|\delta(t_0)\|\, e^{-\lambda(t-t_0)}$ while $\|\delta\| \ge 10^{-6}$, which Theorem 2 bounds below by one, and the last two columns give the time after which $\max_i \|\delta_i\| \le \varepsilon_{\text{tol}}$, against T_{dock} of Corollary 1, common to all rows because $\|\delta(t_0)\|$ is. Every run meets both bounds. Stronger coupling speeds convergence without entering the certificate, and the uncoupled case attains the unit rate, so the bound is tight when the interaction is off. Both deadlines hold from t_0 rather than from τ_c, as Lemma 1 supplies.

C. Formation Reconfiguration

Fig. 2 illustrates the swarm transitioning across four distinct configurations: a ring, a filled wedge, a five-pointed star, and an arrow aligned with r. These geometries encompassing convex, non-convex, and filled lattice templates—are admissible because Assumption 1 imposes no structural constraints beyond centering. Consecutive templates are interpolated via minimum-jerk blends $p_i(t) = (1 - b(t))p_i^k + b(t)p_i^{k+1}$ with $b(t) \in [0, 1]$, preserving Eq. (3) identically and providing $\dot{p}_i$

TABLE III

FORMATION CONVERGENCE AND ACCEPTANCE TIME FOR SEVERAL COUPLINGS.

M	$\hat{\lambda}$	$\varepsilon_{\mathrm{tol}} = 0.1$ (s)	$\varepsilon_{\mathrm{tol}} = 0.01$ (s)
Bound Eq. (16), Eq. (17)	1	5.52	7.83
0	1.00	5.06	7.37
$5I_2$	1.14	4.06	6.08
$5I_2$, rotating	1.13	4.34	6.36
$25I_2$	1.56	2.55	3.98
$5I_2 + 5J$	1.13	4.08	6.10

in closed form. Because certificates (4) and (16) are independent of $p_i(\cdot)$, neither convergence guarantee is affected during reconfiguration.

Station allocation is resolved via the Hungarian algorithm [23] by minimizing $\sum_i \|x_i(t_0) - r(t_0) - p^0_{\pi(i)}\|^2$ at t_0 and $\sum_i \|p^k_i - p^{k+1}_{\pi(i)}\|^2$ at each template change. Station paths move synchronously along straight lines relative to r, guaranteeing collision-free virtual stations under sufficient endpoint clearance [24]. In simulation, this assignment yields a minimum inter-agent clearance of 1.50, compared to 0.006 under naive index ordering. While effective in practice, this remains an allocation heuristic; actual agents follow stations within transient bounds δ_i, so separated virtual targets do not strictly prevent physical collisions.

V. CONCLUSION

This letter shows that kinematic swarms can achieve simultaneous trajectory tracking and formation keeping with independent convergence guarantees. The core mechanism is a shared finite-time centroid feedback that dissipates collective formation errors. This provides two network-independent certificates – i) the swarm centroid converges to the reference trajectory in finite time, and ii) individual station errors decay exponentially from the initial instant to meet an explicit acceptance deadline, which are confirmed by numerical simulations. Independent of the network structure, coupling parameters can be tuned strictly for inter-agent spacing without invalidating the arrival time bounds. Furthermore, these guarantees hold under a limited communication radius across switching proximity components. However, these results have certain limitations such as, practical implementation relies on distributed centroid estimation (e.g. dynamic average consensus), with local damping setting the certified exponential rate, and disturbances and actuation limits need to be incorporated.

Declaration on Generative AI Use: The authors used generative AI tools, including ChatGPT and Claude, solely for the purpose of grammar checking and language refinement. All technical content, proofs, analysis, results, and conclusions were developed and verified by the authors.

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